

FITBUDDY

AI Fitness Plan Generator using Gemini Models

BRAINSTORMING & IDEATION PHASE

Project Overview

FitBuddy is an AI-powered fitness planning application designed to generate personalized fitness plans using Gemini models. The main idea is to help users receive a simple and customized workout routine based on their fitness goals, experience level, available time, and preferences.

Problem Statement

Many people want to become fit but do not know how to create a suitable workout routine. Generic fitness plans may not match an individual's goals, schedule, experience, or preferences. Users may also find it difficult to decide which exercises to perform, how often to train, and how to organize a weekly routine.

Need for the Solution

- Users need fitness plans that are easy to understand and follow.
- A plan should consider individual goals instead of giving the same routine to everyone.
- Users may need quick guidance when creating or changing a workout routine.
- AI can help organize user information into a structured and personalized plan.
- The system should present recommendations in a simple and user-friendly format.

Core Idea

The core idea is to combine user-provided fitness information with Gemini-based AI generation to create a personalized fitness plan. FitBuddy can take inputs such as goal, fitness level, workout days, available time, preferred activities, and basic preferences, then generate a structured plan that the user can understand and use.

PAGE 2 – BRAINSTORMING

Brainstorming Objective

The brainstorming stage focuses on identifying user problems, possible features, useful inputs, and ways AI can make fitness planning easier. The ideas are later filtered to form the main concept of FitBuddy.

Key Ideas Generated

- Personalized workout plan generation
- Goal-based fitness planning
- Beginner, intermediate, and advanced plan options
- Weekly workout schedule
- Exercise suggestions based on available time
- Home-workout and gym-workout options
- Simple exercise instructions
- Workout duration recommendations
- Progress-oriented plan updates
- AI-generated fitness guidance
- Easy-to-read daily routine
- User-friendly interface

Possible User Inputs

Input	Purpose
Fitness Goal	Understand what the user wants to achieve.
Experience Level	Adjust routine difficulty.
Workout Days	Build a realistic weekly schedule.
Available Time	Match workout duration to the user's schedule.
Workout Preference	Choose suitable activity types.
Equipment Availability	Suggest exercises that fit available resources.

Brainstorming Outcome

The brainstorming process identified personalization, simplicity, flexibility, and AI-assisted plan generation as the main directions for the project. These ideas form the foundation for the next ideation stage.

PAGE 3 – IDEATION & FEATURE SELECTION

Selected Project Concept

FitBuddy will act as an AI fitness plan generator. The user enters relevant fitness preferences and goals, and Gemini models process the information to generate a structured fitness plan.

Major Features

- **Personalized Plan:** Creates a plan based on the user's selected goals and preferences.
- **Goal-Based Planning:** Organizes workouts around the user's chosen fitness objective.
- **Difficulty Selection:** Supports different experience levels.
- **Weekly Schedule:** Presents workouts in a clear day-by-day format.
- **Time-Based Planning:** Considers the amount of time available for exercise.
- **Exercise Guidance:** Provides understandable descriptions for suggested exercises.
- **AI Generation:** Uses Gemini models to transform user inputs into a structured response.

Why Gemini Models?

Gemini models can be used to understand natural-language user inputs and generate structured responses. In FitBuddy, the model can help convert a user's goals and preferences into a readable fitness plan. The exact model and implementation can be selected according to the project's technical requirements.

Example User Journey

- Step 1 – User opens FitBuddy.
- Step 2 – User enters fitness goal and basic preferences.
- Step 3 – User selects experience level and available workout time.
- Step 4 – FitBuddy prepares the AI request.
- Step 5 – Gemini generates a structured fitness plan.
- Step 6 – The plan is displayed in an easy-to-follow format.

Design Principle

The project should prioritize simplicity. A user should be able to understand what to enter, receive a clear plan, and follow the generated schedule without unnecessary complexity.

PAGE 4 – PROBLEM-SOLUTION FIT & SYSTEM IDEA

Problem-to-Solution Mapping

User Problem	FitBuddy Idea
Do not know how to create a routine	Generate a structured AI-based routine.
Generic plans may not match personal needs	Use user inputs for personalization.
Limited time for exercise	Generate plans around available time.
Different experience levels	Use experience level as an input.
Confusion about weekly scheduling	Display a day-by-day plan.
Difficulty understanding exercises	Provide simple exercise descriptions.

High-Level System Flow

User Input → Input Validation → Prompt/Request Preparation → Gemini Model → Generated Fitness Plan → Formatting → User Display

Initial Functional Requirements

- The system should collect the required user inputs.
- The system should prepare a suitable AI request from the inputs.
- The system should generate a structured fitness plan.
- The generated plan should be readable and organized.
- The system should handle incomplete or invalid inputs appropriately.
- The interface should make the generated plan easy to view.

Safety and Responsible Use

FitBuddy should be presented as an AI planning and guidance tool, not as a replacement for a qualified healthcare or fitness professional. The project should avoid presenting AI-generated information as guaranteed medical or fitness advice. Users with health concerns, injuries, or other conditions should consider professional guidance.

PAGE 5 – IDEATION SUMMARY & EXPECTED OUTCOME

Final Ideation Summary

The brainstorming and ideation process leads to a clear project concept: FitBuddy is an AI Fitness Plan Generator using Gemini models. The system is intended to make personalized fitness planning simpler by converting user requirements into a structured and understandable workout plan.

Expected Benefits

- Provides a convenient starting point for fitness planning.
- Reduces the effort required to organize a workout routine.
- Makes the plan more relevant by using user-provided preferences.
- Uses AI to generate natural-language and structured recommendations.
- Can be designed as a simple and accessible application.
- Creates a practical use case for Gemini models in an AI application.

Future Enhancement Ideas

- Progress tracking and workout history
- Personalized plan regeneration
- Exercise library with demonstrations
- Reminder and notification features
- Meal-planning support as a separate feature
- Dashboard for weekly progress
- Integration with fitness or activity data where technically appropriate
- Multilingual user interaction

Conclusion

FitBuddy combines fitness planning with generative AI to create a practical, user-focused project. The brainstorming stage identifies the main user problems, while the ideation stage converts those problems into features and a system concept. The resulting idea provides a strong foundation for the next project phases, including design, development, testing, and deployment.

Project Name: FitBuddy – AI Fitness Plan Generator using Gemini Models

Phase: Brainstorming & Ideation Phase

Document: 5-Page Project Ideation Document